

AUTOMATING THE FUTURE

Low Code / No Code Automation & Agentic AI

From Simple Triggers to Autonomous Agents

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Recall Quiz: Previous Knowledge



What is an API?

- ✓ A database
- ✓ **A bridge for software communication**
- ✓ A Python library



Python HTTP Library?

- ✓ pandas
- ✓ **requests**
- ✓ numpy



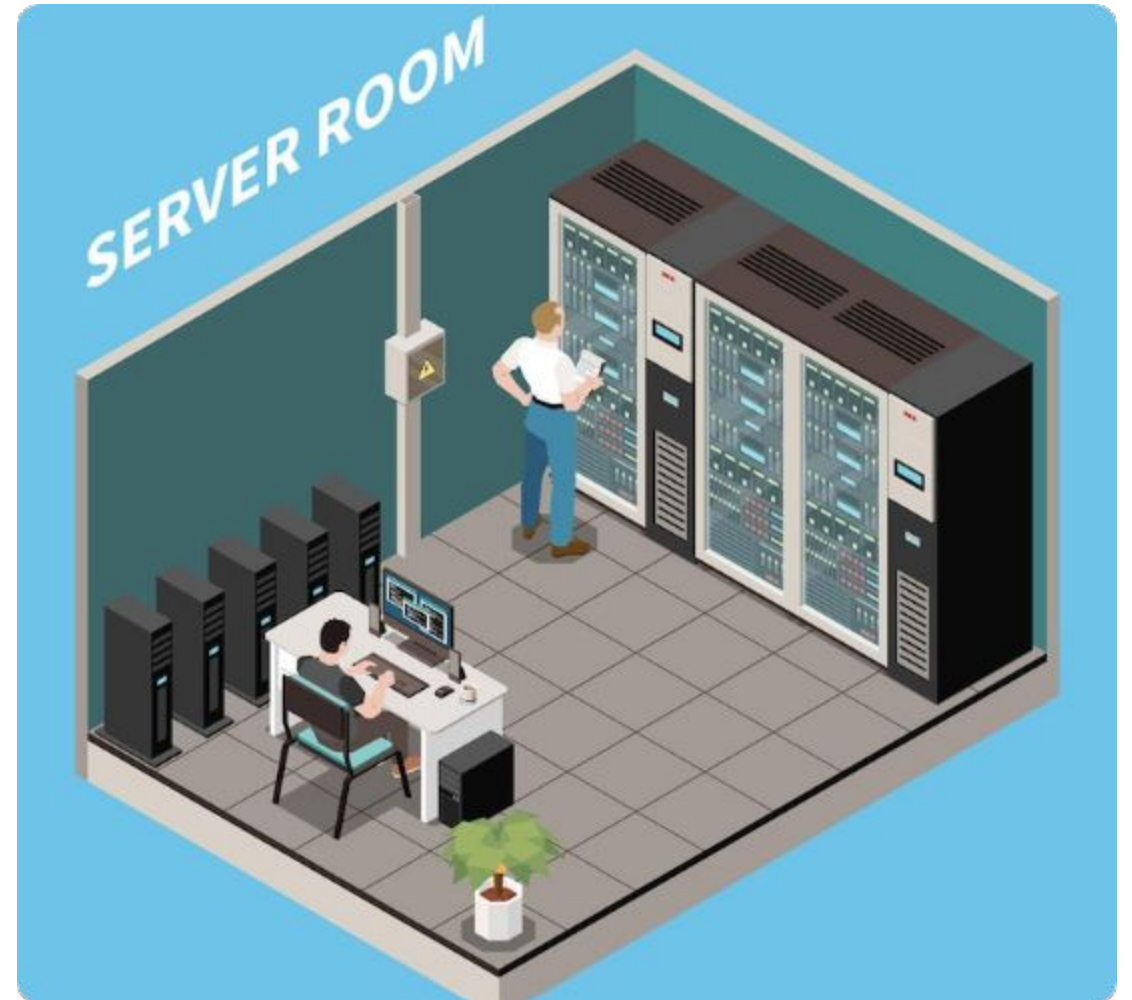
True or False?

"Writing a script from scratch is always the fastest way."

FALSE

Today's Agenda

- ✓ **The Landscape:** Low Code vs. No Code
- ✓ **The Tools:** Zapier, Make, and the power of 
- ✓ **Activity 1:** Thinking in "Triggers" & "Actions"
- ✓ **Beyond Automation:** Introduction to Agentic AI
- ✓ **Practical Demo:** Building an AI Workflow in n8n



The Problem



Drowning in Repetition?

What is the most boring, repetitive task you do every week?

- Copy-pasting data between spreadsheets?
- Sending standard reminder emails?
- Checking server logs manually?

The Reality: We spend 30-40% of our time on "Robotic" work.

What is Low Code / No Code?

Platforms that allow you to build applications and automations using **visual interfaces** (drag-and-drop) instead of writing line-by-line code.



No Code

Zero coding knowledge required. Ideal for business users to connect apps quickly.

Example: Zapier



Low Code

Visual interface but allows custom code (Python/JS) for complex logic and data transformation.

Example: n8n, Make

Why LCNC Matters for Engineers?



Speed to Deploy

Build a prototype in 10 minutes vs. 2 days of writing boilerplate code.



Maintenance

No servers to manage, no crontabs to debug. The platform handles the infrastructure.



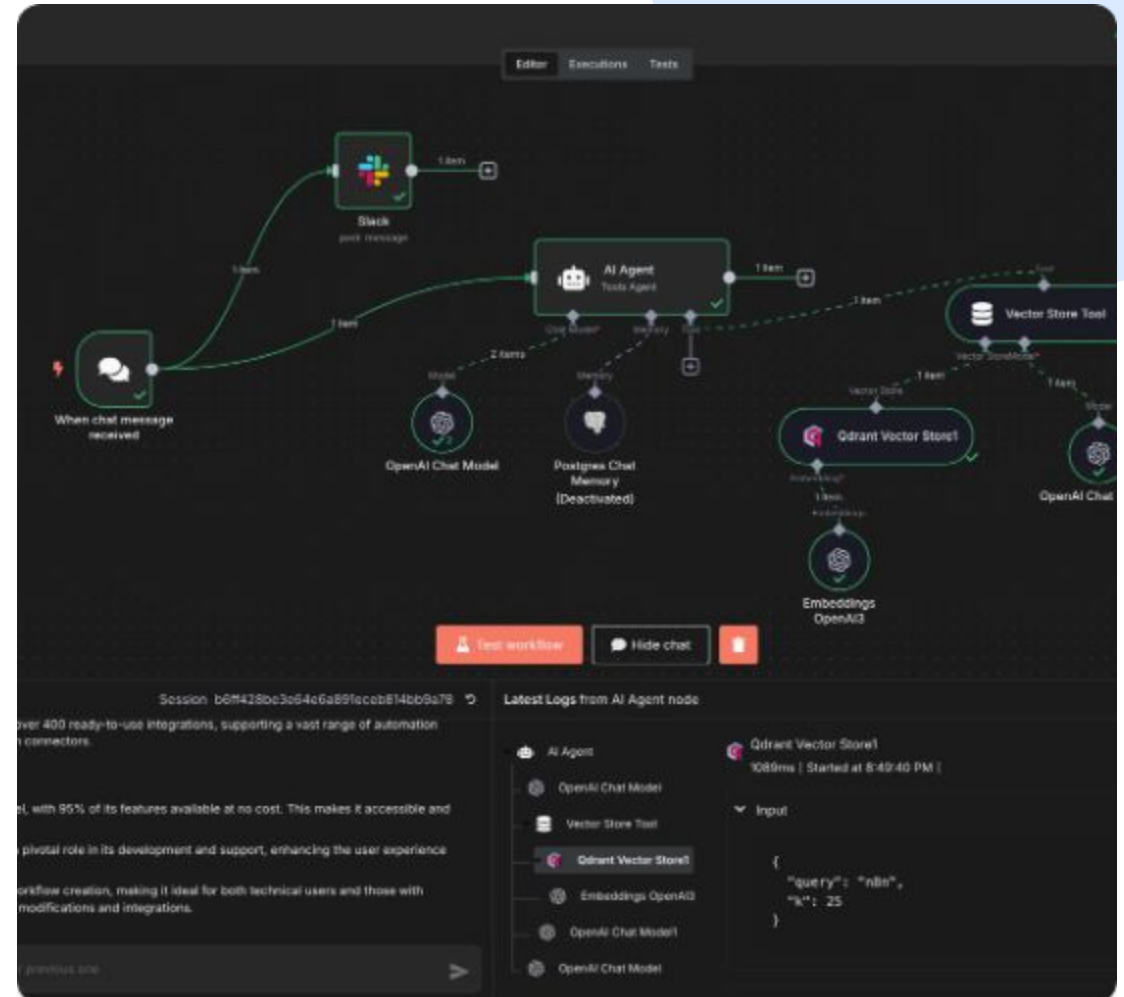
Integration

Pre-built connectors for 8,000+ apps (Slack, Gmail, Notion, OpenAI).

The Developer's Choice: n8n



- ✓ **Best For:** Engineers, AI Agents, Data Sovereignty.
- ✓ **Fair-code:** Self-hostable on your own server (Docker).
- ✓ **Developer Friendly:** Write JavaScript/Python directly inside nodes.
- ✓ **AI Dominance:** Native LangChain integration for Agentic workflows.



Core Concepts of Automation

1. Trigger

The event that starts the flow.

"When a new email arrives..."

2. Filter/Logic

Rules to control the flow.

"Only if subject contains 'Urgent'..."

3. Action

What happens next.

"...send a message to Slack."

The "Big Two": Zapier & Make



Zapier

- ✓ **Best For:** Simplicity & Marketing/Sales teams.
- ✓ **Pros:** Huge library (8,000+ apps), simple "If This Then That".
- ✓ **Cons:** Expensive at scale, limited branching logic.



Make (Integromat)

- ✓ **Best For:** Complex logic & Visual thinkers.
- ✓ **Pros:** Infinite canvas, powerful error handling, cheaper.
- ✓ **Cons:** Steeper learning curve than Zapier.

Tool Comparison



Feature	Zapier	Make	n8n
User Type	Marketing/Sales	Power Users	D Developers
Pricing	High (Per Task)	Medium (Per Op)	L Low / Free (Self-Hosted)
Visuals	Linear List	Bubbles/Graph	N Node Graph
AI Capability	Basic Agents	Intermediate	A Advanced (LangChain)

Scenario: The Bug Report

A user submits a Google Form with a "Bug Report". Design the logic:

- If Priority is "**High**", page the engineer on Slack.
- If Priority is "**Low**", just add it to Trello.
- Regardless of priority, send a "Thank you" email to the user.

Solution: Visual Flow



Standard automation is "Deterministic".

If X happens, do exactly Y.

But what if the input is messy? What if the task requires thinking?

Enter Agentic AI.

What is Agentic AI?

Definition: AI systems that can reason, plan, and act autonomously to achieve a goal.

The Analogy

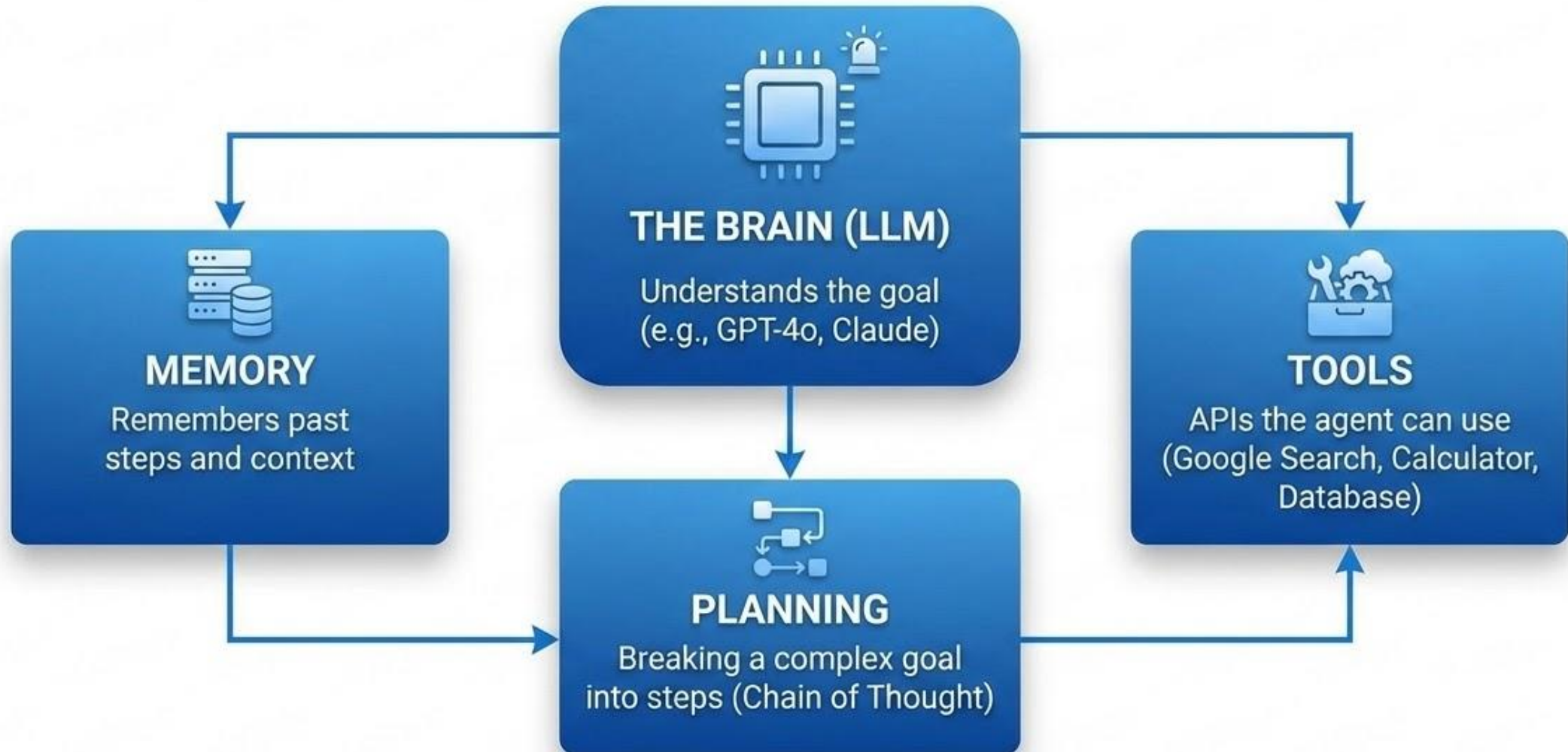
Generative AI (ChatGPT) is a writer.

Agentic AI is a research assistant who has a browser, a phone, and a credit card.

Key Difference

- ✓ **Gen AI:** "Thinker". Produces text/images.
- ✓ **Agentic AI:** "Doer". Uses tools to execute tasks.

THE ARCHITECTURE OF AN AGENT



How an Agent “Thinks” (The Loop)

User Request: “Find Tesla’s stock price and email it to my boss.”



1. Thought

“I need to find the stock price. I have a ‘Yahoo Finance’ tool.”



2. Action

Call ‘Yahoo Finance’ tool for ‘TSLA’.



3. Observation

Tool returns “\$240.50”. Agent thinks: “Now I need to email this.”

Agentic Features in Other Platforms



Zapier Central

Chat with "Bots" that control your Zaps. Instruct them in plain English ("If a lead looks promising, research them on LinkedIn").

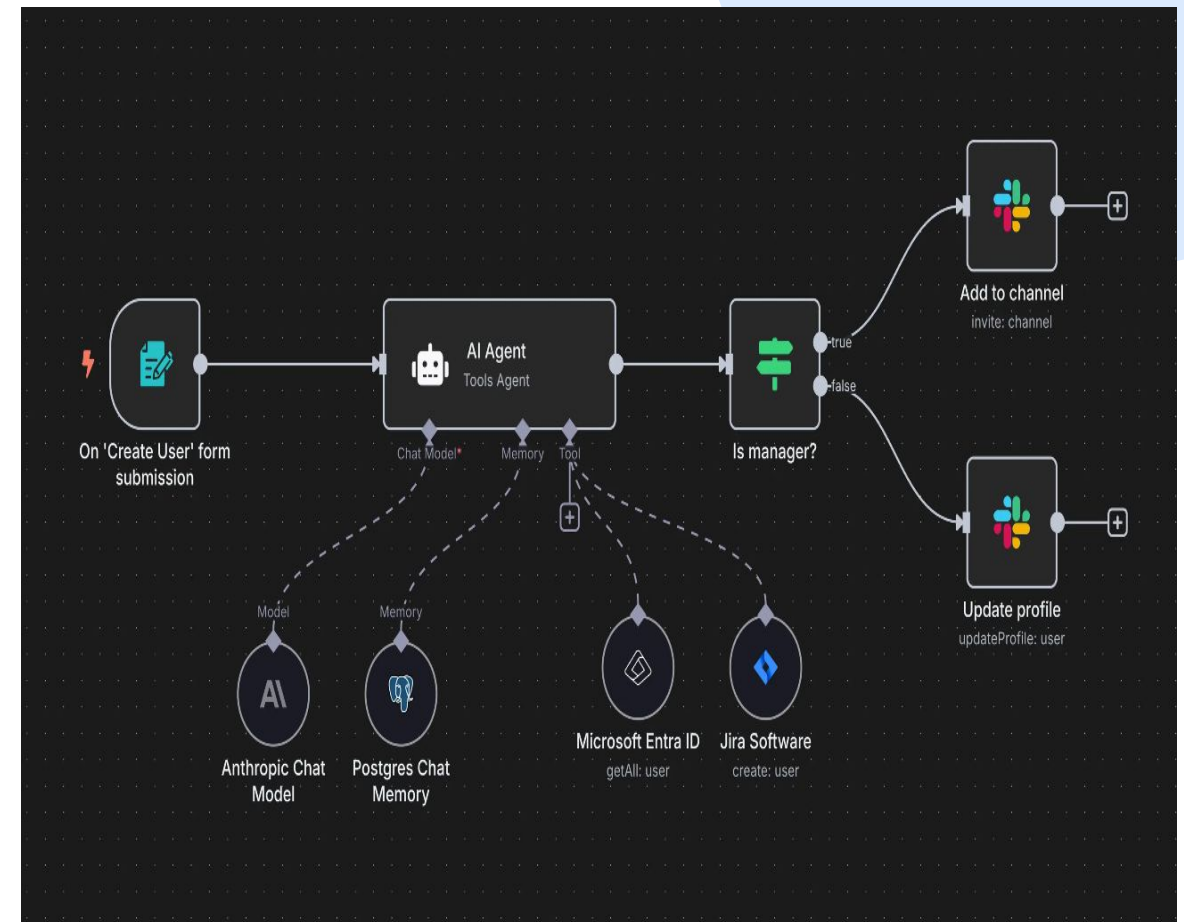


Make AI Assistant

Excellent at orchestration. Use OpenAI modules to analyze data (e.g., extract dates from a contract) and pass it to standard modules.

n8n: The King of Agents

- ✓ **LangChain Integration:** Native nodes for "AI Agent", "Memory", and "Tools".
- ✓ **Flexibility:** Swap models (OpenAI vs Llama 3) easily.
- ✓ **Local AI:** Run agents using Ollama (offline/local models) for privacy.



Checkpoint Activity 2

Task: Define the Tools

Request: "Plan a 3-day trip to Goa for me, find a hotel under ₹5000/night, and add the itinerary to my Google Calendar."

Hint: What APIs does the "Brain" need access to?

Solution: Required Tools



1. Search Tool

(Google Search / Serper)

To find places to visit in Goa.



2. Booking Tool

(Expedia API or Web Scraper)

To check hotel prices and availability.



3. Calendar Tool

(Google Calendar API)

To save the finalized events.

Without these tools, the LLM is just hallucinating!

Human-in-the-Loop (HITL)

The Risk

Agents can make mistakes (buy the wrong ticket, email the wrong person).

The Solution

Pause the workflow for human approval before critical actions.

Workflow Example

- ✓ Agent Drafts Email
- ✓ **PAUSE:** Send approval button to Manager (Slack)
- ✓ Manager Clicks "Approve"
- ✓ Agent Sends Email

Real World Use Case: Data Ingestion

Scenario: You have 1000 PDF invoices in messy formats.

The Agentic Way (n8n + Vision Model)

- ✓ **Trigger:** New file in Google Drive.
- ✓ **Agent:** "Extract Total Amount & Date. Return JSON."
- ✓ **Action:** Insert JSON into SQL Database.



Limitations of Agentic AI



Cost

LLM API calls (tokens) can get expensive quickly.



Latency

Agents are slower than simple code because they have to "think" and wait for generation.



Looping Errors

An agent might get stuck trying to solve a problem forever.

Practical Demo: n8n

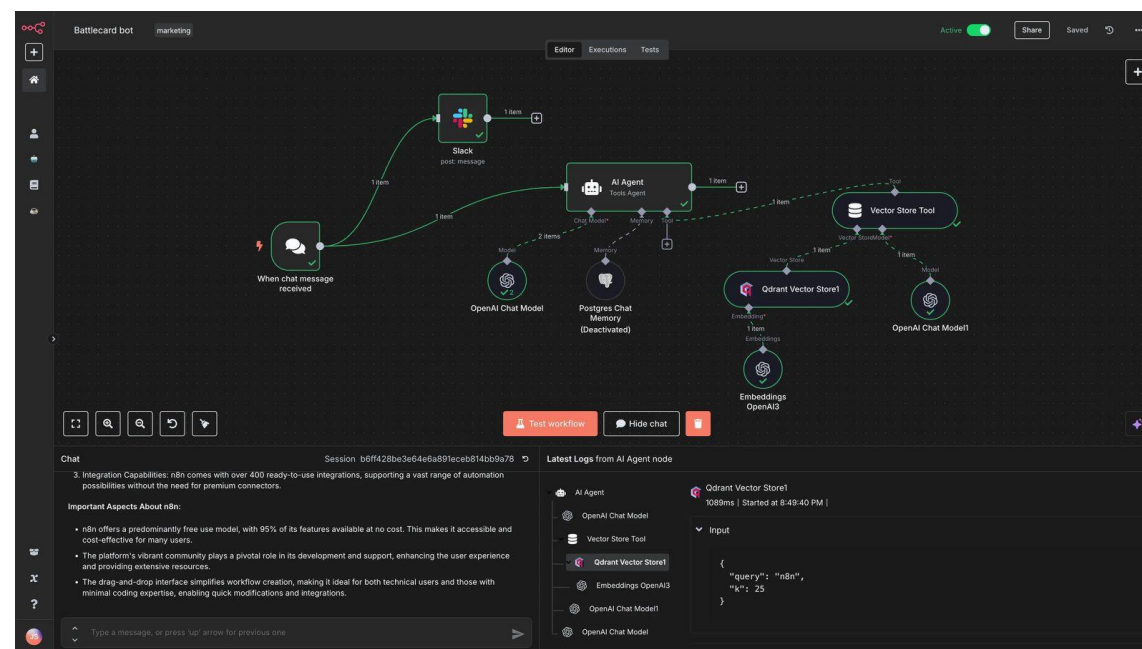
Building an AI Invoice Extractor

We will build a workflow that watches a Google Drive folder, uses AI to read invoices, and saves the data to a database.

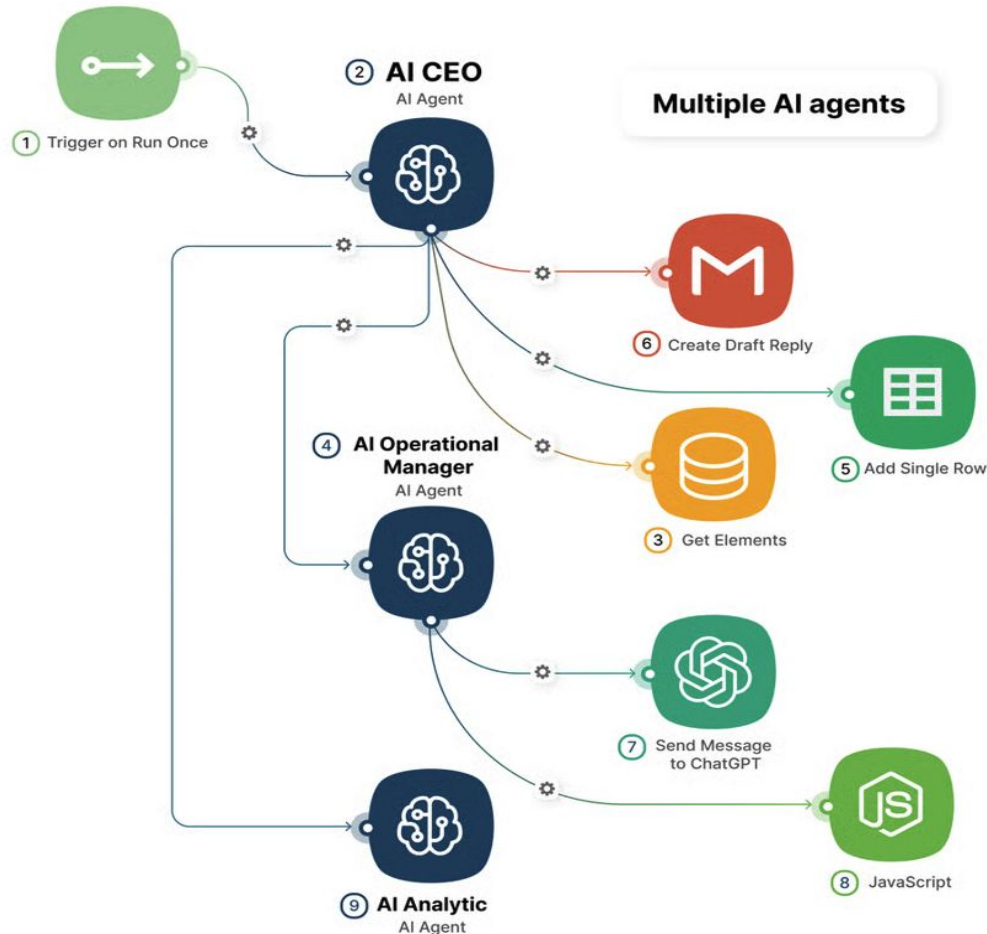
Google Drive Node

We configure this node to watch for **"File Created"** events in a specific folder.

Output: Binary Data (The PDF file)



Step 2: The "Brain" (AI Agent)



Configure the Agent

- ✓ **Model:** GPT-4o (Vision Capable).
- ✓ **Prompt:** "Analyze this image. Extract the 'Total Amount' and 'Date'. Return strictly in JSON format."
- ✓ **Input:** Connect the binary data from Google Drive.

Step 3: The Action

Google Sheets / Database

We map the JSON output from the AI Agent to the columns in our database.

```
Column A ← JSON.date  
Column B ← JSON.total_amount
```





Multi-Agent Systems

One agent for research, one for writing, one for code review—all talking to each other.



Local Agents

Running on your laptop (No data privacy issues, no cloud costs).



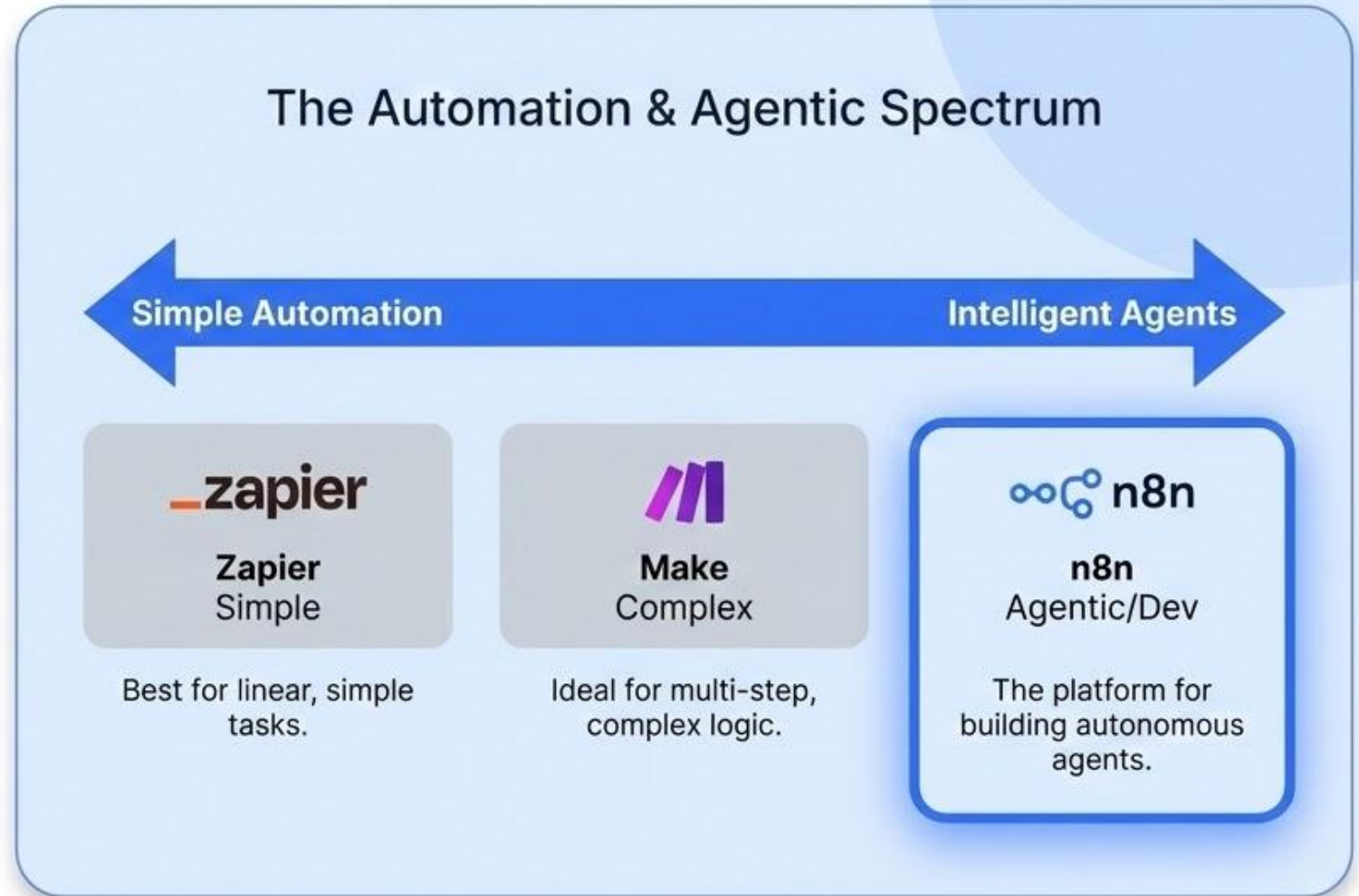
Voice Agents

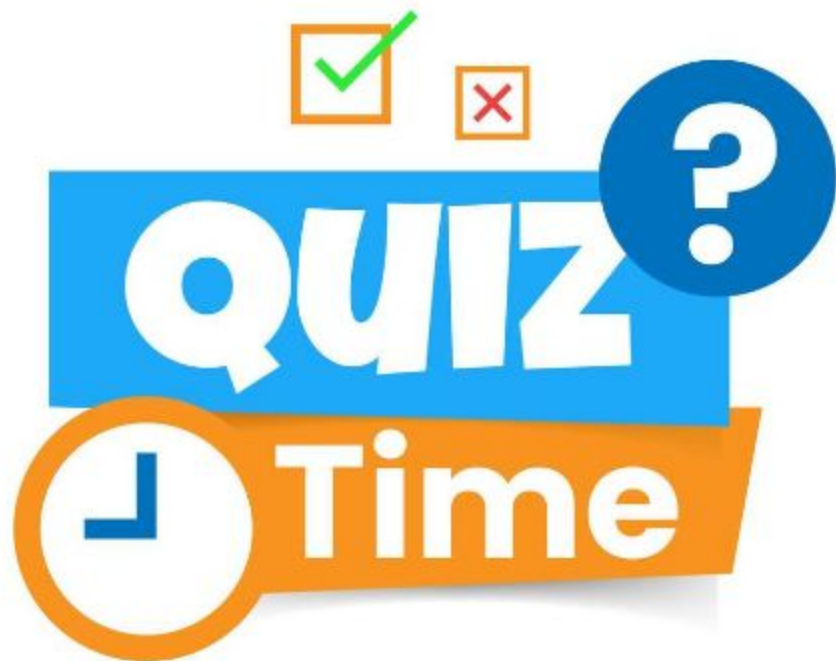
Talking to agents via phone (Voiceflow + Make).

Summary: Key Takeaways & Tool Comparison

Key Takeaways: Low Code vs. Agentic AI

- ✓ **Low Code:**
Democratizes automation.
Moves data between apps without code.
- ✓ **Agentic AI:**
Adds a 'Brain' to the automation. Can plan, reason, and handle unstructured data.





Quiz Time!

Let's test what we've learned.

Question 1

Which tool is best known for being "Self-hostable" and "Developer-friendly"?

- ✓ A) Zapier
- ✓ B) Make
- ✓ C) n8n

Question 2

What is the main role of an LLM in an Agentic Workflow?

- ✓ A) To store the data
- ✓ **B) To act as the "Brain" that plans and reasons**
- ✓ C) To trigger the workflow

Question 3

Why might we need "Human-in-the-Loop"?

- ✓ A) Because AI is too fast
- ✓ **B) To prevent expensive errors or approve sensitive actions**
- ✓ C) To teach the AI how to code

Questions?

Thank you for your attention.

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